

Osama Ahmed Siddiqui

Systems Integration Engineer | Full-Stack AI Developer

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Summary

AI Full-Stack Engineer with 5+ years across **backend** (Python, Flask, FastAPI, Node.js), **cloud** (AWS, GCP, Docker, Frappe Cloud), and **ERP systems** (SAP ABAP & S/4HANA, ERPNext/Frappe, Sage 50, 200, Evolution & X3, Microsoft Dynamics 365, Oracle Fusion). Shipped Nigeria FIRS and Pakistan FBR e-invoicing pipelines for **15+ Tier-1 clients** and published **4 Frappe Marketplace apps**. Additional depth in **AI automation** – RAG/LLM applications (LangChain, OpenAI), computer vision (YOLOv8), n8n workflow automation, and Python ETL at **45K+ daily-record scale**.

Skills

Languages: Python, JavaScript, TypeScript, C/C++, X++, SQL

ERP & Integration: ERPNext, Frappe, Odoo, SAP (ABAP, S/4HANA), Sage (50, 200, Evolution, X3), Microsoft Dynamics 365 (AL, X++), Oracle, AS2/EDI Channels

ML & AI: Supervised & Unsupervised Learning, Neural Networks, Computer Vision, NLP, Time Series, Model Training & Fine-tuning, RAG, TensorFlow, Scikit-learn, LangChain, OpenCV, Ultralytics

Frameworks & Libraries: Flask, Django, FastAPI, Node.js, React, PyQt5, Streamlit, TensorFlow, Scikit-learn, LangChain, OpenCV, Ultralytics

Data & Cloud: ETL Pipelines, AWS, GCP, Docker, MS SQL Server, MariaDB, PostgreSQL, MongoDB, Firebase, Frappe Cloud, Vercel

Tools: Jira, Confluence, ClickUp, Slack, Airtable, n8n, QuickBooks, Git

Work Experience

Cryptware Systems LTD

Apr 2026 – Present

System Integration Expert

Remote | NG

- Architected cross-ERP payload transformations and IRN/QR injection flows as an **Access Point Provider** to enable Nigeria **FIRS/NRS e-invoicing** compliance across **16+ Tier-1 clients** spanning SAP, Oracle, Sage, and Dynamics 365 ecosystems.
- Delivered Sage ERP e-invoicing integrations across four product lines for Nigeria FIRS compliance: **Sage 50** (Pervasive ODBC + 32-bit Python/Flask dashboard) for **Proton Security Services**; **Sage 200** (ODBC middleware) for **Genesis Group**; **Sage Evolution** (SQL middleware); and **Sage X3** (cloud-based SData 2.0 REST API) for **Swift Oil** and **Chorus Energy** – eliminating manual FIRS submissions across all entities.
- Developed **ZCL_FLICK_API_INTEGRATION ABAP** classes, a custom **ZFIRS** transaction, and **RFC/STRUST** setup to migrate **Dabur LTD** and **Fidelity Bank PLC** SAP pipelines onto Cryptware APIs across DDE/DQE/DPE landscapes.
- Authored **AL extensions** and **X++ customizations** for **Microsoft Dynamics 365** to deliver NRS/FIRS e-invoicing, covering **Prime Atlantic Group's 9 entities**, **VINCI Energies**, and Fidelity Bank PLC.
- Provisioned **AS2 middleware** and **EDI channel** integration for **Pernod Ricard** to bridge their global ERP with the Nigeria access point, enabling automated B2B invoice exchange.
- Migrated **Keystone Bank's Oracle** e-invoicing pipeline from Flick Network APIs to Cryptware Systems APIs, consolidating vendor dependency and cutting integration maintenance overhead.
- **Stack:** SAP ABAP, S/4HANA, Sage 50, Sage X3, Microsoft Dynamics 365 (AL, X++), Oracle, Pervasive ODBC, Python, Flask, AS2/EDI, REST APIs.

Delivery Devs

Oct 2024 – Present

Technical Consultant

Hybrid | PK

- Spearheaded a **Frappe CRM** rollout and **SQL Server data warehouse pipeline** (Daxium + Sage ERP sources) for **AF Compressors (Belgium)** to centralize sales analytics across a multi-country territory, enabling territory-based opportunity tracking and unified dashboards.
- Led an **ERPNext Finance/HR/Operations** implementation for **Orion Group LLC**, devising custom workflows and analytics to consolidate three departmental systems into one ERP and reduce reporting cycle time.
- Released a Tourism Management application on Frappe with **S3** and **Google Drive** backup pipelines and a **global-passport OCR module**, automating **70% of operational workflows** and removing manual passport entry.
- Authored an **FBR E-Invoicing Frappe app** to bring multiple Pakistan-based clients into compliance with the FBR digital invoicing mandate.
- Open-sourced **three Frappe Marketplace apps** – **Exchange Rate Synchronization**, **ZKTeco biometric check-in integration**, and **Cargo Management** – with the Cargo Management app raising operational efficiency by **80%** for adopting logistics clients.
- Tuned **InnoDB** and **Redis** configurations at Falcon-i Tracking to resolve persistent ERPNext timeout errors, restoring stable production performance.
- **Stack:** Frappe, ERPNext, Python, MariaDB, SQL Server, Redis, AWS S3, Google Drive API, OCR.

Upwork

Freelance AI, ERP & Integration Engineer

Jan 2025 – Present

Remote | DE, US

- Delivered **Slack approval flows**, **two-way Airtable sync**, and **Google Drive backup integrations** on **ERPNext** for **Motley Terpz** (US-licensed cannabis manufacturer) to automate cross-platform inventory and approval workflows, eliminating manual data entry between systems.
- Redesigned **ERPNext UI/UX**, dashboards, and process visualizations for cannabis tolling operations and stood up production/staging environments with SSL, enabling the functional team to scale manufacturing and dispatching workflows independently.
- Built a **Facebook-Ads-to-ERPNext lead pipeline** via **n8n** and configured **multi-company Chart of Accounts** for **LabPro Pharma** to unify sales and dispatching across **3 country entities** (Nigeria, Cameroon, DRC), automating lead capture from paid social.
- Constructed a full-stack **sports prediction app** (PyQt5, Flask, Streamlit, NeonSQL) using **Monte Carlo simulations** and **Random Forest classification** with **Stripe payment integration**, achieving **70% prediction accuracy** across NBA, NHL, MLB, and NFL with live odds tracking.
- Engineered a **Bolt.eu RBAC integration** for **theurbandrive.de** to enforce role-based access across **30,000 drivers** and **60 in-house employees**, securing fleet-management data and eliminating cross-team credential exposure.
- Deployed **KETO GPT**, a **Retrieval-Augmented Generation (RAG) LLM application** on Vercel, to deliver personalized keto recommendations and nutrition-score tracking for end users.
- Programmed an **AI-powered crypto trading bot** (Tkinter GUI) combining SuperTrend, Golden/Death Cross, Bollinger Bands, and Funding Rate indicators to automate **Binance order execution**, removing manual signal-tracking from live trading.
- Shipped **ResumeAI**, a resume-enhancement platform with region-specific optimization, **ATS scoring**, customizable templates, and an **AI chat editor**, to help job seekers tailor resumes per market and improve ATS pass-through rates.
- **Stack**: Python, Airtable API, Google Drive API, n8n, Slack API, PyQt5, Flask, Heroku, Streamlit, NeonSQL, LangChain, OpenAI, Stripe, Vercel, Binance API.

Lookflock

Data Engineer & AI Developer

Oct 2021 – Oct 2024

Remote | UK

- Constructed **Python ETL pipelines** across **55 fashion brands** to extract **45,000+ daily data points**, streamlining data handling and reducing process downtime by **30%**.
- Established a **CI/CD pipeline** for automated testing and deployment to replace manual release cycles, reducing deployment time by **50%** and increasing release reliability.
- Devised a **product validation API** and **user-activity logging system** to power personalized content delivery, boosting user engagement by **30%**.
- Built a **computer vision pipeline** using **OpenCV** and **Ultralytics** for automated fashion product tagging and visual similarity search, reducing manual categorization effort by **60%**.
- Developed an **NLP-based recommendation engine** trained on user interaction history and product metadata, improving click-through rates by **22%** across personalized feeds.
- Architected a **real-time data warehouse** on **Firebase** with **Redis** caching layer to support analytics dashboards, cutting query response times by **40%**.
- Engineered a **Brand Manager Portal** enabling fashion brands to self-manage their storefronts, publish posts, schedule social content, and run **targeted boost campaigns** to amplify reach across the platform's user base.
- Built an **admin panel** for internal ops teams to oversee brand onboarding, moderate content, manage platform-wide settings, and monitor brand activity and engagement metrics in real time.
- Refined social interaction features and UI flows to reduce friction in the engagement loop, increasing daily active users by **20%**.
- **Stack**: Python, Flask, TensorFlow, Scikit-learn, OpenCV, Ultralytics, ETL, CI/CD, REST APIs, PostgreSQL, Firebase, Redis, Docker.

Education

FAST – National University of Computer and Emerging Sciences (NUCES)

Bachelor of Science in Computer Science

Aug 2020 – Jun 2024

CGPA: 3.43 / 4.0

Three-time Dean's List Honoree. Relevant coursework: Applied Machine Learning, Cloud Computing (AWS), IS Audit & Control (CISA Domains 1 & 2), Recommendation Systems, Deep Learning for Perception, Information Processing Techniques (C# & .NET).

Final Year Project

Fly Free Eye: Drone-Powered Unconstrained Object Monitoring

- Augmented collected drone imagery datasets to address class imbalance, increasing training sample size by **40%**.
- Benchmarked multiple **computer-vision architectures** (CNN, Deep CNN, YOLOv8 variants) across accuracy, latency, and edge-compute metrics to select **YOLOv8-s** as the production model.
- Deployed YOLOv8-s on a **DJI Mavic** with a **Google Coral** and **Raspberry Pi** edge stack to classify objects across 10+ classes in real time, achieving **87.08% classification accuracy**.
- **Stack**: DJI SDK, OpenCV, Kotlin, TFLite, Ultralytics, Flask.

Certifications

- **IBM Data Science Professional Certificate** – Coursera
- **Full Stack Frappe Developer** – Frappe School
- **Neural Networks & Deep Learning** – DeepLearning.AI
- **LLM & RAG** – freeCodeCamp
- **SQL Intermediate** – DataCamp | **Power BI & Excel** – DataCamp | **Problem Solving Intermediate** – HackerRank